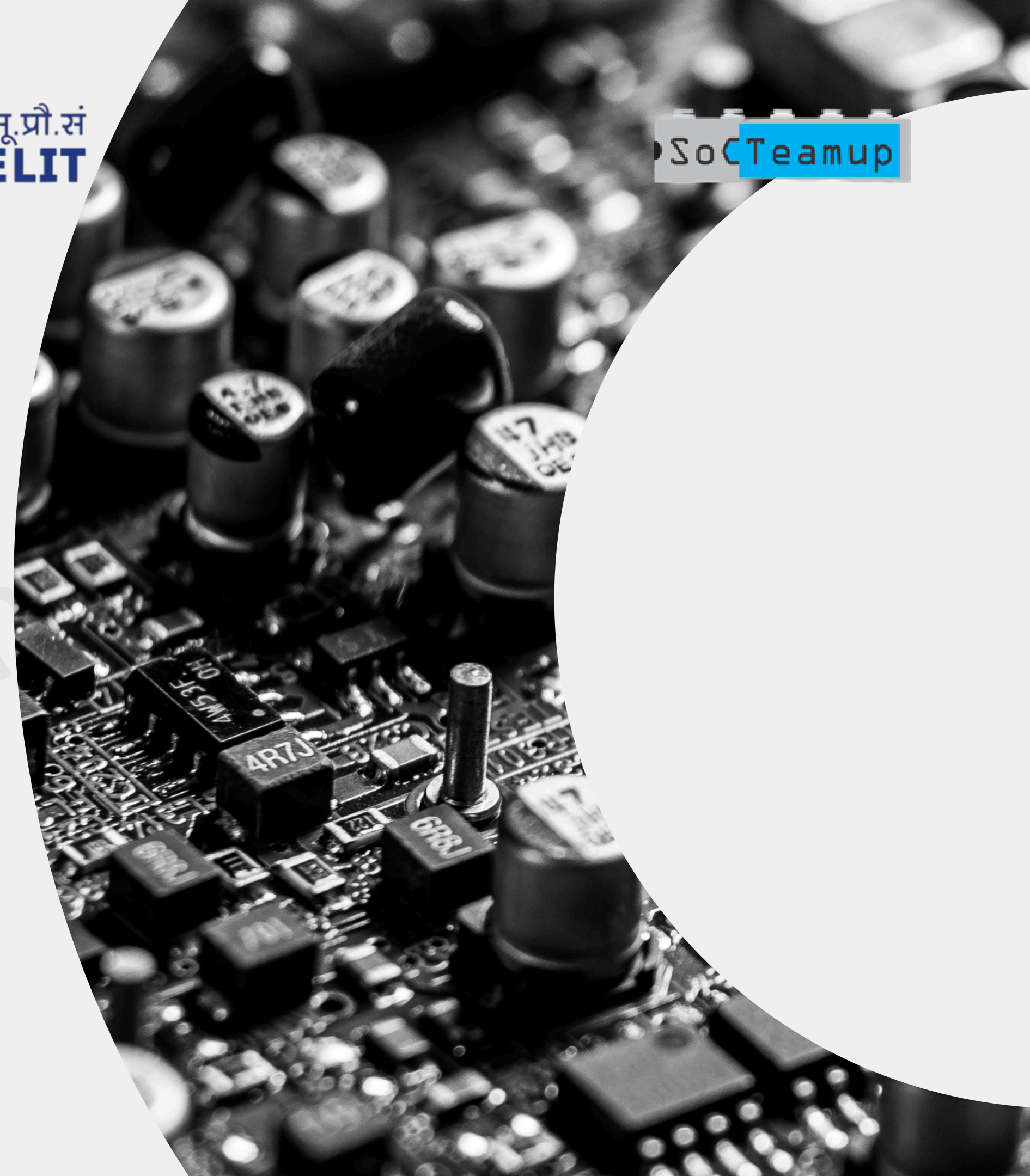


Low-power Considerations during Physical Design





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➔ Introduction: Power in ICs

- Power is a critical constraint in modern VLSI systems, especially for portable and battery-powered devices.
- Excessive power → more heat, leakage, and reliability issues.
- During physical design, power optimization is as important as timing and area.

Equation for total power:

$$P_{total} = P_{dynamic} + P_{static} + P_{short-circuit}$$

Goal: Reduce each component without affecting performance.

➔ Sources of Power Consumption

Power Type	Source	Dependency
Dynamic Power	Switching of logic signals	Frequency (f), Voltage (V), Capacitance (C)
Static Power	Leakage through transistors	Process, Temperature, V_t
Short-Circuit Power	Overlap of conduction in PMOS/NMOS	Transition activity

Dynamic power dominates during switching, while leakage dominates in idle states.

➔ Dynamic Power Explained

Dynamic Power Equation:

$$P_{dynamic} = \alpha \cdot C_L \cdot V^2 \cdot f$$

Where:

- α : Activity factor (0–1)
- C_L : Load capacitance
- V : Supply voltage
- f : Clock frequency

To reduce it:

- Lower voltage (V)
 - Reduce switching activity (α)
- Use clock gating or smaller cells

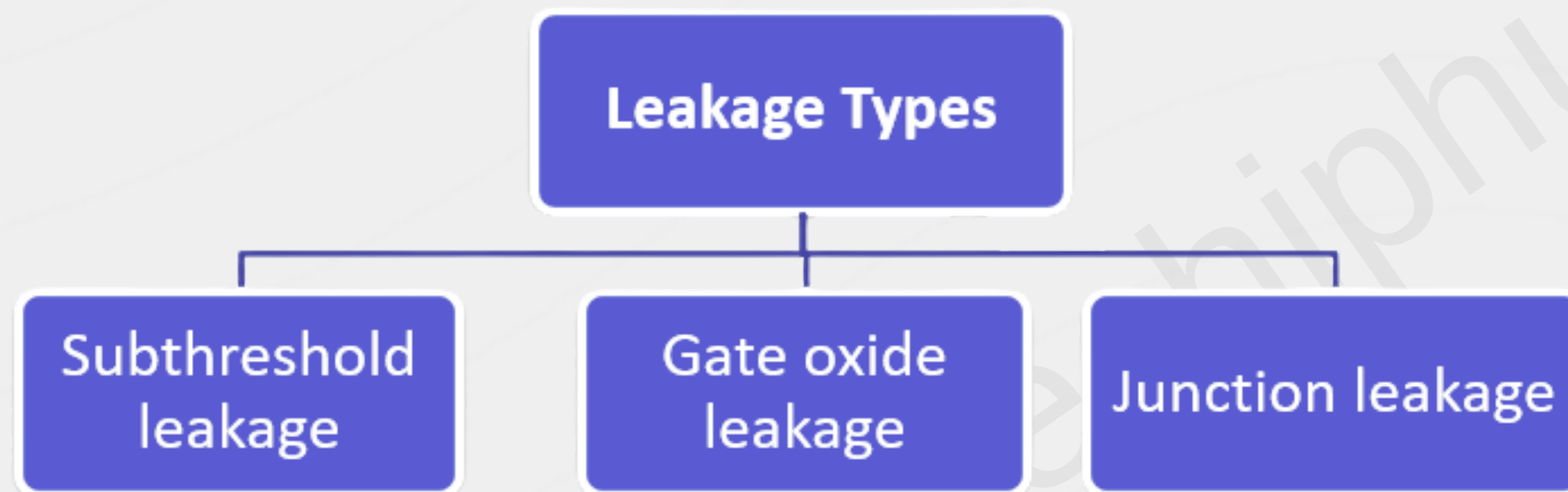
Example Calculation:

$$\text{If } \alpha = 0.5, C = 10pF, V = 1V, f = 500MHz:$$
$$P = 0.5 \times 10^{-11} \times 1^2 \times 5 \times 10^8 = 2.5mW$$

➔ Static Power (Leakage Power)

Leakage power comes from **transistor leakage currents** even when idle.

$$P_{static} = I_{leak} \times V_{DD}$$



Trends:

As transistors shrink, leakage increases exponentially — thus **low-leakage cell libraries** and **power gating** become critical.

➔ Low-Power Design Goals

Goal	Description
Minimize Dynamic Power	Use clock gating, lower switching, reduce voltage
Minimize Leakage	Use multi-Vt, power gating, high-Vt cells
Optimize Performance	Balance timing vs power
Ensure IR/EM Safety	Avoid excessive voltage drop or metal heating

Power, performance, and area (PPA) must be co-optimized during design.

Clock Gating (Dynamic Power Reduction)

Concept: Disable clock signal to inactive modules to save dynamic power.

Example Verilog:

```
module clock_gating (  
    input wire clk, en,  
    output wire gated_clk  
);  
    assign gated_clk = clk & en;  
endmodule
```

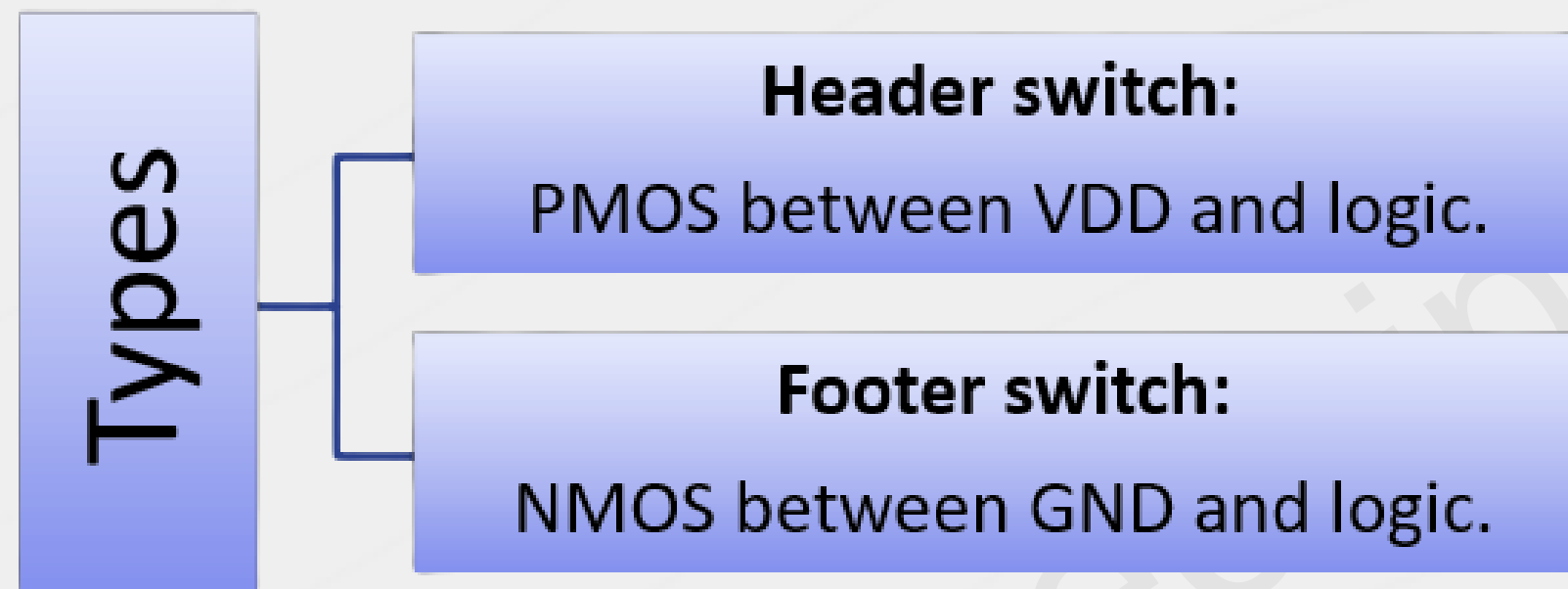
Explanation:

- When $en=0$, gated_clk stops toggling → no unnecessary switching.
- In real designs, this is replaced by an **Integrated Clock Gating (ICG)** cell to avoid glitches.

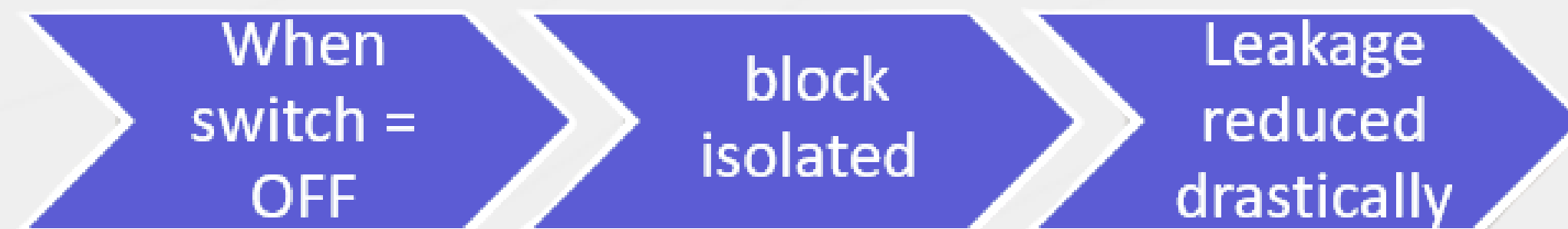
➔ Power Gating (Static Power Reduction)

Concept: Cut off power to idle blocks using power switches (sleep transistors).

Used in: Standby modes of SoCs and mobile processors



Example: [VDD] ----> [Power Switch] ----> [Logic Block]



Multi-Vt Optimization

Concept: Use different threshold voltage (V_t) cells for trade-off between speed and leakage.

Cell Type	Speed	Leakage
Low-Vt	Fast	High
High-Vt	Slow	Low

Strategy:

- Timing-critical paths → **Low-Vt**
- Non-critical paths → **High-Vt**

Supported by synthesis and place & route tools (Yosys + OpenLane example).

Multi-Voltage Domains

Concept: Divide design into voltage islands.

Each region runs at an optimal voltage to balance speed and power.

Domain	Voltage	Function
Core	0.9V	Performance-critical
Peripherals	0.7V	Low-speed logic

Isolation Cells: Prevent signal corruption between voltage domains.

Level Shifters: Adjust signal levels when crossing domains.

Command Example (UPF):

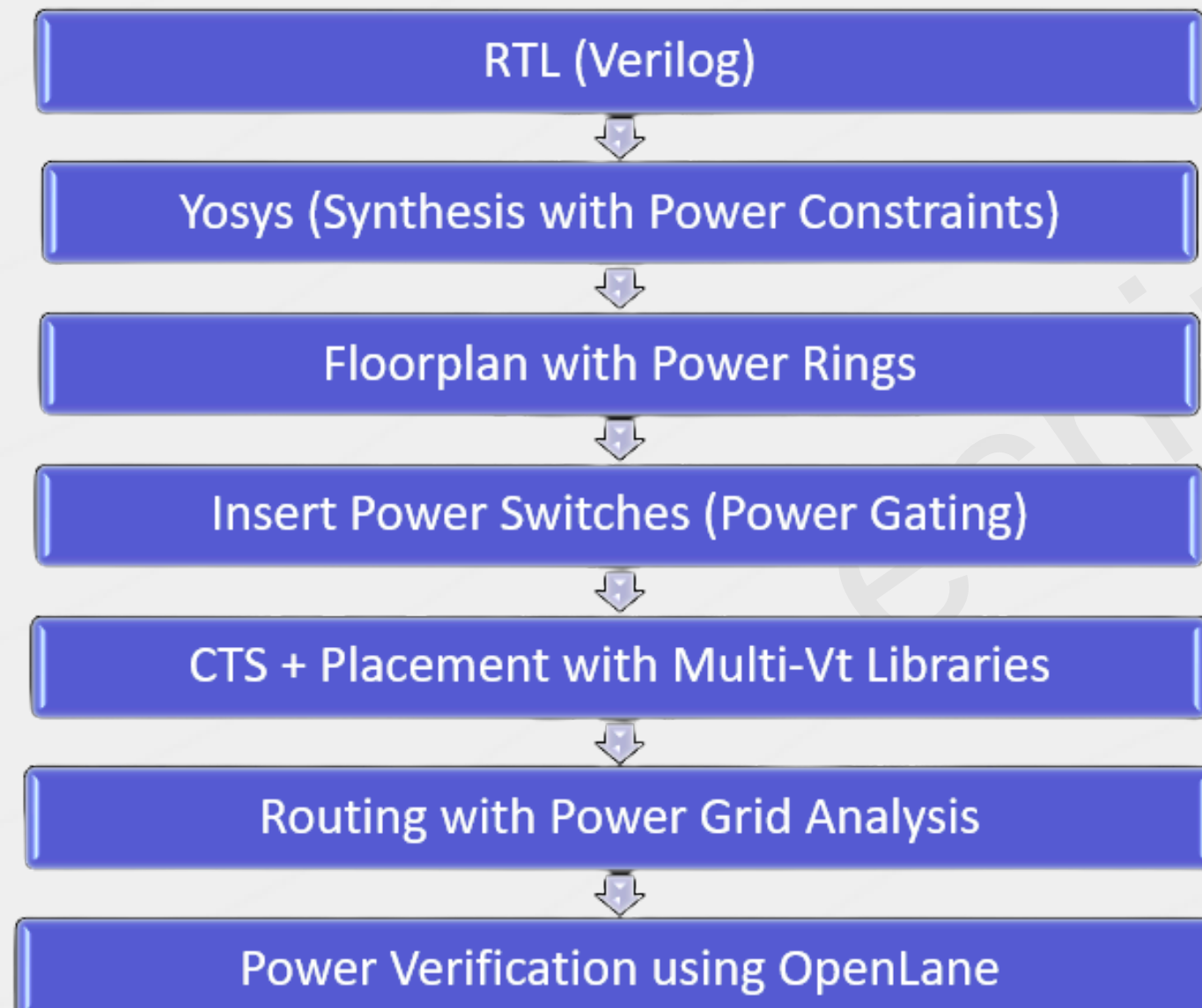
```
create_power_domain CORE -elements {u_core}  
set_domain_voltage CORE 0.9
```

Power Grid and IR-Aware Design

- Ensure sufficient **power distribution network (PDN)** to avoid IR drop.
- Poor PDN design can negate low-power benefits due to unstable voltage delivery.
- Wider metal rails, redundant vias, and decoupling capacitors improve stability.

Tools like OpenRCX and Magic can extract resistive parasitics for IR-awareness.

➔ Example Power-Aware Flow (Conceptual)



Power Intent Description (UPF Basics)

- **UPF (Unified Power Format)** describes power domains, voltages, and power states.
- Defines how **isolation**, **retention**, and **switching** are applied.
- Portable across tools and stages.

Simple Example:

```
create_power_domain PD1 -elements {block1}  
set_domain_supply_net PD1 -primary VDD1 -ground VSS  
add_isolation_rule -from PD1 -to PD2 -isolation ISO_CELL
```

Purpose: Maintain consistent power definitions from RTL to GDSII.

→ Key Design Trade-offs

Technique	Advantage	Trade-off
Clock Gating	Saves dynamic power	Minor control overhead
Power Gating	Reduces leakage	Adds wake-up delay
Multi-Vt	Balances speed/leakage	Requires careful timing closure
Multi-Voltage	Scales power vs performance	Needs level shifters
UPF	Portable intent	Adds verification complexity

Rule: Combine multiple low-power methods for maximum benefit.

➔ Summary and Best Practices

Summary

Power optimization starts early and continues through layout.

Use clock gating for dynamic power, power gating for leakage, multi-Vt and multi-voltage for flexibility.

Ensure a robust PDN for IRN stability.

Best Practices

Keep clock activity factor low via gating.

Use multi-Vt synthesis for leakage control.

Run post-layout IR/EM analysis before signoff.

Balance PPA (Power, Performance, Area) using iterative optimization.

Thank You

echiphu

